Discovering the Lyrical Drivers of TikTok Virality

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Research Question

The knowledge we wish to extract from the data is which lexical and structural features of current song lyrics drive a song to be TikTok viral [1]. Leading our main research question to be:

> **Main Research Question:** *Which lexical, structural, and stylistic characteristics of song lyrics distinguish TikTok-viral tracks from standard commercially successful music?*

- **Subquestion 1 (Lyrical Simplicity):** Are TikTok-viral lyrics significantly simpler (lower lexical diversity) and more repetitive (higher compressibility and duplicate lines) than non-viral commercial hits?
- **Subquestion 2 (Stylistic Markers):** Which stylistic patterns (such as conversational pronoun usage, line brevity, Part-of-Speech distributions, and rhyme density) best characterize TikTok-viral songs?

Motivation & Approach

Recent studies demonstrate that song lyrics have become progressively simpler and more repetitive over the past five decades [3], yet they do not identify the modern media mechanisms driving this shift. We hypothesize that short-form video platforms such as TikTok reinforce and accelerate this trend, as songs must capture listener attention and support lip-sync or visual trends within 15-second snippets. As Alaminos Fernández [2026] show, musical virality on TikTok is not random, but emerges from a structured convergence of platform-native aesthetics and algorithmic dynamics. While Prevedello et al. [2024] showed that song popularity can be predicted from lyrics using black-box Large Language Model embeddings, predictive accuracy alone does not yield interpretable domain knowledge about *why* specific lyrics resonate on short-form platforms.

We approach this by combining lexicometric text mining with interpretable stylometric modeling. By using classification models to understand the lyrics, we leverage feature-group ablation and SHAP explanations to isolate, rank, and explain the specific linguistic markers that distinguish viral from non-viral lyrics.

Data Description

Our base corpus is the `genius-lyrics-cleaned` dataset [2], filtered to English songs released between 2020 and 2025. To avoid confounding platform virality with obscurity, we curate a balanced corpus of 6,000 tracks via the Spotify Web API: 3,000 TikTok-viral songs from trending

[1] Tiktok virality is a concept defined as "blowing up quickly and engaging followers in a way that is surprising", see https://www.forbes.com/sites/johnbbrandon/article/how-to-go-viral-on-tiktok/

playlists and 3,000 non-viral commercially successful songs from mainstream charts (excluding tracks in the viral set). Tracks are linked to the Genius corpus via strict normalization of artist and title, deduplicating remixes to prevent data leakage.

Validation

We combine internal and external validation:

- **Internal Validation:** We evaluate feature differences using 95% Confidence Intervals and Chi-Square (χ^2) tests. As standard machine learning evaluation, we assess our classifier on an 80/20 stratified test split (with 5-fold cross-validation) using Accuracy, Precision, Recall, and F_1-score, cross-referencing SHAP rankings with lexicometric metrics.
- **External Validation:** We validate findings against independent sources without relying on metrics: (1) corroborating extracted trends with musicological literature on lyrical simplification [3, 1], and (2) performing qualitative close reading on prototypical songs to verify that identified repetitive hooks correspond to the actual lyric snippets trending in TikTok videos.

References

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[3] Emilia Parada-Cabaleiro, Maximilian Mayerl, Stefan Brandl, Marcin Skowron, Markus Schedl, Elisabeth Lex, and Eva Zangerle. 2024. Song lyrics have become simpler and more repetitive over the last five decades. *Scientific Reports* 14 (03 2024). doi:10.1038/s41598-024-55742-x

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